SOKOINE UNIVERSITY OF AGRICULTURE
DEPARTMENT OF AGRICULTURAL ENGINEERING

ABE 715: COMPUTER PROGRAMMING

Date Given: Friday, 13/02/2026 Submission Date: Friday, 06/03/2026
Student Name: _____ Reg. No: _____

INSTRUCTIONS:

1. This Term Paper constitutes the final project for ABE 715 and spans 3 weeks.
2. There are FOUR project options. You must coordinate in class to ensure each student selects a different project.
3. Geospatial Data: For all projects requiring administrative boundaries, you must use official District or Ward shapefiles from the National Bureau of Statistics (NBS) Tanzania.
4. Method Comparison: Every project requires a rigorous comparison between your designed Python methodology and an existing alternative method (e.g., specialized open-source tools or validated spreadsheet models).
5. Deliverables: Functional Python scripts, a technical report (8–12 pages) formatted for journal submission, and publication-quality visuals (High DPI, 300+).

Project Option 1: The "Digital Dakawa" Rice Scheduler

Focus: Data Science, APIs, and Soil Water Balance
Location: Dakawa Irrigation Scheme, Mvomero (Lat: -6.4422, Lon: 37.5586).

Problem Background: The Dakawa scheme faces significant water productivity challenges due to rigid pumping schedules that ignore actual evapotranspiration (ET_o) rates. Over-irrigation causes nitrogen leaching and high electricity bills, while delayed irrigation during the reproductive phase severely impacts rice yields. Farmers need a dynamic tool that uses real-time satellite weather data to track soil water depletion and trigger irrigation only when the management allowed depletion (MAD) is reached.

Task: Develop a Python-based Dynamic Water Balance Model.

(a) Use the NASA POWER API to retrieve 10 years of daily climate data for the Dakawa coordinates.

(b) Implement the FAO-56 Penman-Monteith equation to compute daily Reference Evapotranspiration (ET_o).

(c) Program a soil water depletion loop for a rice crop cycle: $D_i = D_{i-1} - (P_i - RO_i) - I_i + ET_{c,i} + DP_i$.

(d) Comparison: Compare your Python-calculated irrigation triggers against results from the FAO CROPWAT 8.0 software. Discuss any discrepancies in ET_c estimation.

(e) Visuals: Generate a publication-quality time-series graph showing daily depletion levels relative to Field Capacity and Permanent Wilting Point.

Project Option 2: Geospatial Suitability for Avocado in Njombe

Focus: GIS, GeoPandas, and Raster Analysis
Location: Njombe District Council (Lat: -9.3333, Lon: 34.7667).

Problem Background: The commercial expansion of avocado in the Southern Highlands is often hindered by poor site selection. Avocados are sensitive to poor drainage (requiring slopes $> 2\%$) and are prone to erosion on steep terrain (slopes $> 30\%$). Extension officers lack a rapid automated tool to identify suitable land within specific NBS-defined Ward boundaries using topographical and climatic raster data.

Task: Develop a Multi-Criteria Evaluation (MCE) Suitability Tool.

(a) Integrate NBS Ward maps with SRTM 30 m Digital Elevation Models (DEM) and WorldClim rainfall/temperature rasters.

(b) Calculate Slope, Aspect, and Rainfall-Temperature suitability indices using Rasterio and NumPy.

(c) Comparison: Compare your Python grid results against a parallel analysis performed using the Google Earth Engine (GEE) Python API or JavaScript environment. Discuss differences in data pixel alignment and processing efficiency.

(d) Visuals: Create a colorful layout map showing Suitability Classes overlaid with NBS Ward boundaries, including scientific legends and scale bars.

Project Option 3: Makutapora Basin Groundwater Recharge Estimator

Focus: Data Science, Hydro-meteorology, and Numerical Integration
Location: Makutapora Wellfield, Dodoma (Lat: -5.9875, Lon: 35.7533).

Problem Background: Dodoma relies on the Makutapora Basin for its water supply. Climate change and variable rainfall patterns in semi-arid Tanzania make it difficult to estimate the annual groundwater recharge. Engineers need to automate the calculation of "Effective Rainfall" that escapes evapotranspiration and surface runoff to reach the aquifer. This requires integrating daily rates over an entire hydrological year.

Task: Develop a Python-based Annual Recharge Simulator.

(a) Retrieve climate data for the Dodoma basin via the NASA POWER API or CHIRPS dataset.

(b) Program a soil-moisture-accounting model to estimate daily deep percolation.

(c) Use Simpson's 1/3 Rule to numerically integrate daily recharge rates into annual total volumes ($m^3/year$).

(d) Comparison: Compare your Python model results against a standard Thornthwaite-Mather Water Balance calculated manually or via a validated Spreadsheet model.

(e) Visuals: Plot cumulative recharge vs. cumulative rainfall for a 5-year period to identify aquifer response patterns.

Project Option 4: Lower Moshi Irrigation Potential Mapping

Focus: GIS, Distance Analysis, and Surface Modeling
Location: Lower Moshi Irrigation Scheme (Lat: -3.4211, Lon: 37.3789).

Problem Background: The Lower Moshi area has significant potential for expansion, but development is constrained by proximity to primary canals and the Rau River. Identifying expansion zones requires analyzing topographical accessibility and soil drainage. A programmatic tool is needed to find areas that are within 500 m of water sources and have a slope of 1–5% within specific NBS-defined Wards.

Task: Develop a Geospatial Accessibility & Potential Tool.

(a) Use NBS Ward maps to define the study area. Extract river and canal lines from OpenStreetMap (OSM) using Python.

(b) Implement a Proximity (Buffer) analysis and a Slope analysis using GeoPandas and Rasterio.

(c) Comparison: Compare your Python-calculated potential area (in hectares) against the results from the FAO Global Agro-Ecological Zones (GAEZ) platform for the same region.

(d) Visuals: Generate a colorful heatmap representing "Irrigation Potential Scores" across the Lower Moshi wards, ensuring NBS ward boundaries are clearly visible.

END OF PROJECT OPTIONS